

Sunwoo Hong

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RESEARCH INTERESTS

I work on **closing the gap between what people do easily and what models still cannot**: people reason instead of matching patterns, learn from very little, and act in a physical world they understand. I pursue this through **reasoning**, **efficient learning**, and **world models for embodied agents**, by finding the assumption that holds a model class back and replacing it.

Keywords: reasoning, world models, embodied AI, diffusion language models, data-efficient learning

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)	Seoul, South Korea
M.S. in Artificial Intelligence, Graduate School of AI	<i>Feb 2027 (expected)</i>
Advisor: Prof. Se-Young Yun	
Seoul National University (SNU)	Seoul, South Korea
B.S. in Electrical and Computer Engineering	<i>Feb 2024</i>
Sejong Academy of Science and Arts	Sejong, South Korea
Inaugural Student Council President; Government-designated academy for gifted students	<i>Feb 2018</i>

PUBLICATIONS

*: equal contribution (random order), †: corresponding author, C: conference, W: workshop, U: under review, P: in preparation, D: domestic

- [C2] S. H. Kim*, **Sunwoo Hong***, Y. Choi, C.-H. Chao, S.-Y. Yun†, and R. G. Krishnan†. Pivot-SD: Efficient Self-Distillation for Masked Diffusion Language Models. **Empirical Methods in Natural Language Processing (EMNLP), Main Conference**, 2026. **Oral presentation**. [\[project\]](#)
- [C1] S. H. Kim*, **Sunwoo Hong***, H. Jung, Y. Park, and S.-Y. Yun. KLASS: KL-Guided Fast Inference in Masked Diffusion Models. **Neural Information Processing Systems (NeurIPS)**, 2025. **Spotlight (top 3.2% of submissions)**. [\[pdf\]](#) [\[code\]](#)
- [U1] **Sunwoo Hong***, S. H. Kim*, C.-H. Chao, Y. Choi, S.-Y. Yun, and R. G. Krishnan. Logical Equivariance in Diffusion Language Models. Under review. Earlier version presented at **ICML Workshop** on Foundations of Deep Generative Models, 2026. [\[pdf\]](#)
- [W1] C. Park, **Sunwoo Hong**, K. Kim, Y. Park, A. W. Olson, and A. Pesaranghader†. Route-Guided CBF-QP Repair for Safe Vision-Language-Action Models. **RSS Workshop** on Trustworthy Embodied Foundation Models, 2026. [\[pdf\]](#) [\[project\]](#) [\[code\]](#)
- [P1] **Sunwoo Hong**. Do World Models Need Clocks? In preparation for submission to CVPR 2027.
- [D1] **Sunwoo Hong**, Y. Kim, S. Jung, J. Oh, and S.-Y. Yun. Memory Reduction via Ternary Quantization in Reinforcement Learning. Korea Institute of Military Science and Technology (KIMST) Conference, 2025.

RESEARCH EXPERIENCE

Graduate Research Assistant, OSI Lab, KAIST AI	<i>Feb 2025 – Present</i>
Advisor: Prof. Se-Young Yun [page]	
– Building a world model that predicts the next physical event, its timing, and the post-event state for each candidate action, instead of rolling out fixed-interval frames [P1]. – Proposed a training-free sampler for masked diffusion models that unmask tokens whose predictions are stable under KL divergence: up to $2.78\times$ faster generation while improving over greedy decoding [C1].	

Visiting Researcher, University of Toronto & Vector Institute*Jan 2026 – Jul 2026**Advisor: Prof. Rahul G. Krishnan* [\[page\]](#)

- Showed that a masked diffusion language model can be post-trained on under 4% of its tokens, outperforming full-sequence fine-tuning and online RL runs that take six to eight times the wall-clock time [C2].
- Introduced logical equivariance, an evaluation of whether correctness survives answer-preserving rewrites of a problem. A post-training recipe raises the all-view correctness of LLaDA-8B from 47.7% to 86.1% [U1].

Research Collaborator, LG Electronics Toronto AI Lab*Jan 2026 – Jul 2026**Mentor: Dr. Ali Pesaranghader* [\[page\]](#)

- Co-developed a repair module that adds route guidance to a CBF-QP safety filter, so a vision-language-action policy can rejoin its task after avoiding an obstacle [W1].

Graduate Research Assistant, MLML Lab, KAIST School of Computing*Feb 2024 – Jan 2025**Advisor: Prof. Sungjin Ahn* [\[page\]](#)

- Designed a compact 1D tokenizer that compresses video into a short sequence of global tokens, to model scene-level dynamics in video world models.
- Studied why block-slot object-centric models fail to disentangle factors such as shape and color, with out-of-distribution protocols that measure compositional generalization.

Undergraduate Researcher, AISys Lab, SNU*Jan 2023 – May 2023**Advisor: Prof. Jinho Lee (B.S. thesis)* [\[page\]](#)

- Built a distributed training framework for knowledge distillation, using pipeline parallelism to reduce memory use and pipeline bubbles.

Research Intern, Human-Centered Computer Systems Lab, SNU*Jun 2022 – Sep 2022**Advisor: Prof. Youngki Lee* [\[page\]](#)

- Built a server-driven adaptive video streaming pipeline for deep learning inference, reducing bandwidth by up to 70% while maintaining detection accuracy.

Undergraduate Research Program (URP), SNU*May 2021 – Dec 2021**Advisor: Prof. Kyomin Jung* [\[page\]](#)

- Co-developed a GAN-based model that transfers a sign-language interpreter’s motion, extracted as skeletons, onto the speaker in a video, so that the speaker appears to sign.

AWARDS AND HONORS

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| • Oral Presentation, EMNLP 2026 | <i>Oct 2026</i> |
| • Excellence Selection, GARL 2026 Challenge (National AI Research Lab, Korea) [U1] | <i>Jul 2026</i> |
| • Spotlight (top 3.2% of submissions), NeurIPS 2025 | <i>Dec 2025</i> |
| • Full Scholarship, Choi Hyuk-Young Scholarship Foundation | <i>Mar 2019 – Present</i> |
| • Minister of Education Award (1st place), national STEAM R&E research competition | <i>Nov 2015</i> |

TEACHING EXPERIENCE

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| • Teaching Assistant, Geometric Deep Learning (AI810), KAIST | <i>Spring 2025</i> |
| • Teaching Assistant, Reinforcement Learning (CS672), KAIST | <i>Fall 2024</i> |
| • Teaching Assistant, Freshman Seminar: Writing (E51.101), Seoul National University | <i>Fall 2021</i> |

ACADEMIC SERVICES**Reviewer** ACL Rolling Review (August 2026), ICLR 2027**SKILLS****Languages:** Python, C++, C **Frameworks:** PyTorch, JAX, TensorFlow**MILITARY SERVICE**

- Republic of Korea Air Force, Sergeant (honorable discharge); information systems specialist *Aug 2019 – May 2021*